

Flywheels

from zero to wheel

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today



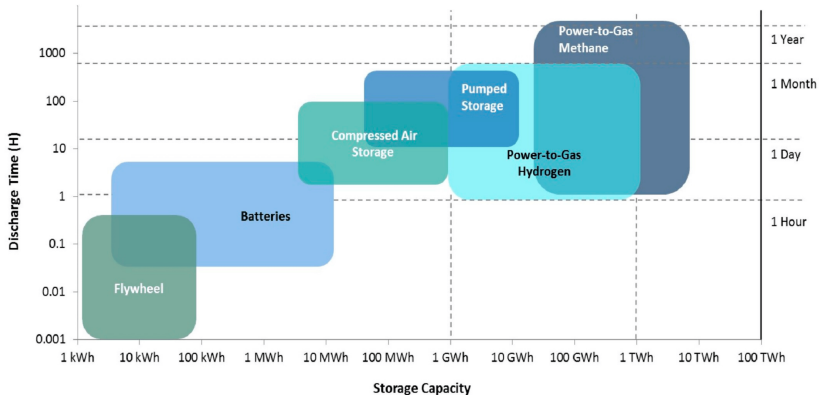


Figure: Comparison of discharge time vs capacity of energy storage technologies (source: doi.org/10.3390/en9090674)



Physical characteristics

moment of inertia: $J = \int_m r^2 dm$

rotational energy: $E_{kin} = \frac{1}{2} J \omega^2$



Specific energy

$$\frac{E}{m} = K \frac{\sigma}{\rho}$$

- ▶ K – shape factor [dimensionless]
- ▶ σ – tensile strength [Pa]

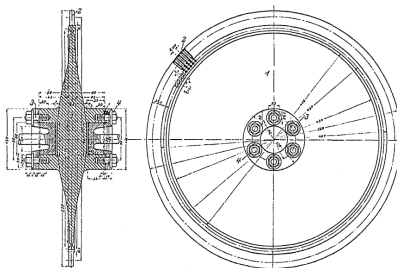


Figure: Constant stress turbine disc built by De Laval at the end of XIX century (source: doi.org/10.1007/BF01556455)

thanks for attention



https://github.com/falcotinnunculus/przezrocza/blob/wladca/red_energy/flywheel.pdf

